

Chemistry 3A

Introductory General Chemistry

Experiment 8a

Determination of Gas Constant R

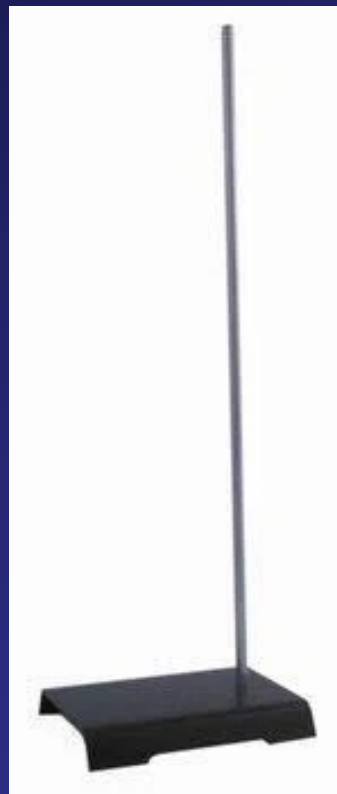
Introduction

- The metal **magnesium** (**Mg**) solid will react with acids (in this experiment, **hydrochloric acid HCl**) to produce **hydrogen** (H_2) **gas** as shown in this reaction
$$\text{Mg} (s) + 2 \text{HCl} (aq) \rightarrow \text{H}_2 (g) + \text{Mg}^{2+} (aq) + 2 \text{Cl}^- (aq)$$
- The **ideal gas law equation** **$PV = nRT$** will be used to determine the value of **R** , the **ideal gas law constant**
- You will use the **mass** (in **grams**) of **Mg** and its **molar mass** value, along with knowing **volume V** , **pressure P** , and **temperature T** , to compute the **R** value

Equipment You Will Use

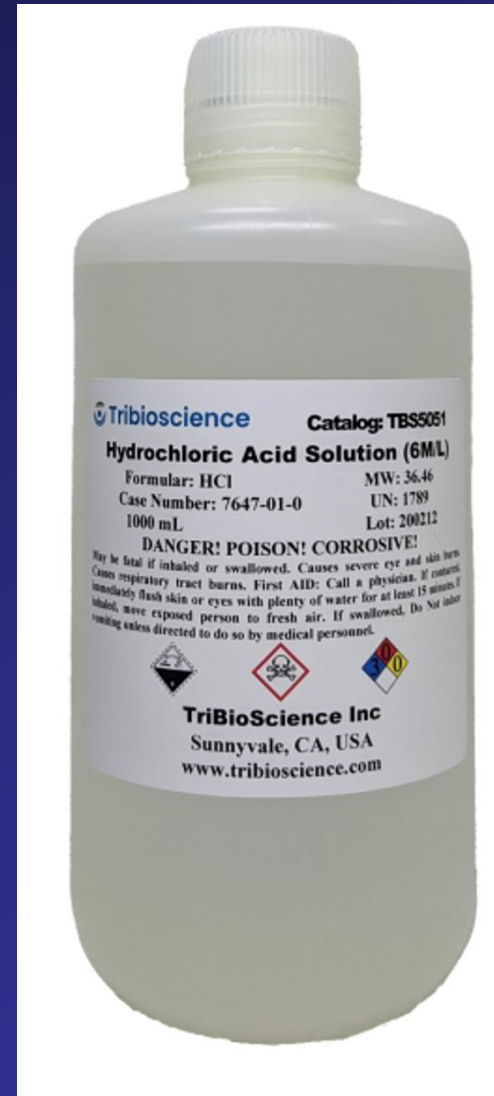


400, 600 or 800 mL beaker



**50 mL gas
collection tube
(eudiometer)**

Consumables



Pre-Lab Preparation/Questions

A pre-lab quiz has been made for you

THIS IS AN EXTRA CREDIT quiz that is worth 15% (30 points) of the value of a lab report (200 points). I suggest you do the quiz if you have been wanting extra credit points for the course

It will help you understand and complete this laboratory report as well

The quiz is at this URL

CRC Handbook

- Your lab manual wants you to look up a reference for the **water vapor** at a **particular temperature**
- The manual mentions an **authoritative reference** called the **CRC Handbook of Chemistry and Physics**
- Fortunately, this reference is available online at [this link](#)

- This is a PDF which may load directly in your browser

- If you did a search for “water vapor” or “vapor pressure” inside the PDF, you would find the table you are looking for. The “vapor pressure” search worked in this case.

VAPOR PRESSURE OF WATER FROM 0 TO 370° C							
This table gives the vapor pressure of water at intervals of 1° C from the melting point to the critical point.							
REFERENCE							
Haar, L., Gallagher, J.S., and Kell, G.S., <i>NBS/NRC Steam Tables</i> , Hemisphere Publishing Corp., New York, 1984.							
<i>t</i> /°C	<i>P</i> /kPa	<i>t</i> /°C	<i>P</i> /kPa	<i>t</i> /°C	<i>P</i> /kPa	<i>t</i> /°C	<i>P</i> /kPa
0	0.61129	53	14.303	106	125.03	159	602.11
1	0.65716	54	15.012	107	129.39	160	617.66
2	0.70605	55	15.752	108	133.88	161	633.53
3	0.75813	56	16.522	109	138.50	162	649.73
4	0.81359	57	17.324	110	143.24	163	666.25
5	0.87260	58	18.159	111	148.12	164	683.10
6	0.93537	59	19.028	112	153.13	165	700.29
7	1.0021	60	19.932	113	158.29	166	717.83
8	1.0730	61	20.873	114	163.58	167	735.70

- You can use this table, but you might have to convert to other pressure units (keep this in mind)
- For your citation: write in the URL. You can even add at the end of the URL the string “#page=<page number you find>” and it will go straight to the page

Procedure

1. **Magnesium ribbon: DO NOT HANDLE THE RIBBON WITH BARE HANDS!!! ONLY HANDLE Mg ribbon with gloves!** Otherwise it might irritate your hands
2. Clean the **Mg** ribbon by lightly rubbing with steel wool. It may become “shiny” as an oxide outer layer is removed
3. Put weighing paper on the balance, zero it, and put the ribbon on the balance: you are looking for a target mass of about **0.05 g (50 mg)**. If the **mass > 0.05 g**, estimate carefully how much ribbon to break off, break it, then re-weigh it to get the final mass you will use. Record the mass (all the digits from the balance) in your lab manual

Procedure

4. You need to break off the ribbon so as not to lose too much mass: you will need to tie your Mg ribbon to fishing line (~15 cm long)
5. Set up the **clamp** on a **ring stand** and **beaker** as shown in figure below. Fill the **beaker** halfway with **water**

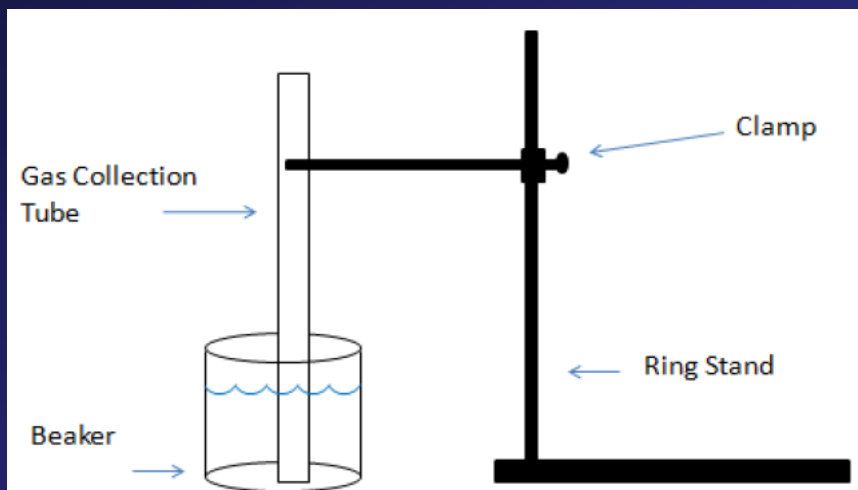
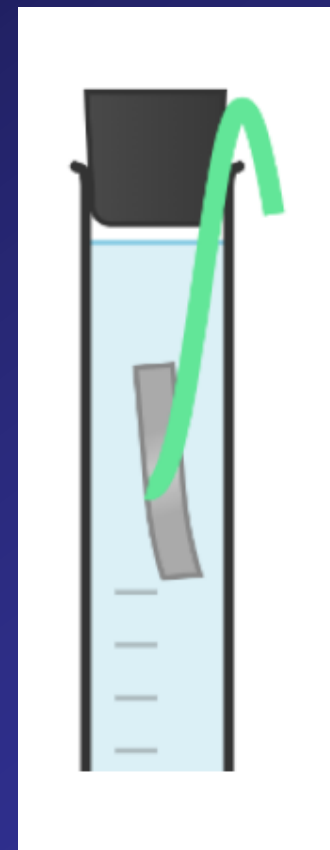


Figure 1: Gas Collection Set-up



Procedure

6. Using a small (10 mL) **graduated cylinder**, pour **8-10 mL** of **6 M** (molar) **hydrochloric acid (HCl)** into the gas collection tube (eudiometer)
7. For this step of adding **DI water**, I suggest you first tip the **eudiometer** tube at an angle so that the water comes in down the side of the wall of the eudiometer tube. **VERY SLOWLY AT FIRST** add water to the eudiometer to avoid **MIXING** the water with the acid. As you add more water, the mixing effect will lessen but still add it slowly. Bring the level of the water to the brim of the eudiometer tube, this time holding the tube quite vertically.
8. Put the **Mg ribbon** in the water **JUST INSIDE THE TUBE**. Some of the **fishing line** should be **OUTSIDE** the tube.
9. Place a **rubber stopper** in the tube, making sure it holds the fishing line

Procedure

10. Put your GLOVED finger over the hole in the rubber stopper: INVERT the tube with your finger still over the hole and immerse your gloved hand—finger still over the hole—into the beaker of water. Quickly but carefully put the tube in the clamp and hold it. You can remove your finger but only with tube with stopper immersed in beaker water.
11. Use a thermometer to record beaker water temperature to nearest 0.1°C . Also get the classroom barometric pressure
12. The reaction is done WHEN
 - There are no bubbles forming
 - The Mg ribbon is not visible
 - The headspace gas is not increasing

You might want to ask your instructor if he thinks the reaction is complete
13. Record volume of gas produced.

The Fishing Line: Why?



Why the magnesium ribbon is tied with fishing line

1. Controlled reaction start

- If you just dropped the Mg ribbon directly into the HCl at the bottom of the eudiometer, the reaction would begin *immediately* while you're still trying to invert and set up the tube.
- By suspending the Mg ribbon higher up in the water column, the acid doesn't touch it until the apparatus is inverted and the acid diffuses downward. This gives you time to get the setup stable before gas starts evolving.

2. Preventing premature mixing

- Remember, the HCl is carefully layered at the bottom, with water above it. If the Mg ribbon fell straight into the acid layer, the reaction would be violent and uncontrolled.
- The fishing line holds the ribbon in place above the acid, so the reaction begins gradually as the acid diffuses upward into the water and reaches the Mg.

3. Keeping the Mg in place

- Without the line, the ribbon might float, stick to the glass, or move unpredictably as bubbles form.
- The line ensures the Mg stays suspended in the same spot, making the reaction more reproducible and the gas collection more reliable.

4. Safety and measurement accuracy

- A sudden burst of H_2 could blow liquid out of the stopper hole or cause splashing.
- By slowing the initial contact, the fishing line helps the reaction proceed at a manageable rate, so the gas volume can be measured accurately.

Details of Reaction

Step-by-step sequence after inversion

1. Before inversion

- Tube is filled: ~8–10 mL of 6 M HCl at the bottom, carefully layered water above.
- Mg ribbon is suspended on fishing line, held near the top of the column, away from the acid.
- No reaction yet, because the Mg hasn't touched the acid.

2. Inversion into the beaker of water

- Tube is inverted and placed in the water bath.
- The stopper hole allows displaced liquid to escape as gas is later produced.
- The Mg ribbon is now hanging "below" the acid layer (since the tube is upside down), but still separated by water.

3. Diffusion begins

- HCl slowly diffuses upward into the water column.
- As the acid front reaches the Mg ribbon, the reaction starts gradually:



Details of Reaction



Why the fishing line matters in this timeline

- It **delays the start** of the reaction until the setup is stable.
- It **controls the position** of the Mg so the acid reaches it gradually.
- It **prevents chaos** (Mg floating, sticking, or reacting too violently).

Details of Reaction

4. Gas evolution

- Hydrogen gas bubbles form on the Mg surface.
- The bubbles rise and collect at the top of the inverted tube, displacing water out through the stopper hole.
- The fishing line keeps the Mg in place so it doesn't float away or stick to the glass.

5. Reaction continues

- Acid keeps diffusing, sustaining the reaction until the Mg is consumed.
- The gas volume increases steadily, pushing more water out.
- Students can then read the gas volume against the scale on the eudiometer.

6. Completion

- Once all Mg is reacted, gas collection stops.
- The measured gas volume (corrected for water vapor pressure and temperature) is used to calculate moles of H_2 , and from that, the experimental gas constant R .

Clean Up

- Lift the tube and remove the stopper to allow the contents of the tube to drain in the beaker water
- Take the beaker—which contains the strong HCl acid—to the sink. Use the NaHCO_3 solid at your bench and add very small amounts of it to the beaker to neutralize the acid. When it stops producing gas, it is neutralized. Pour the contents in the sink while running the tap water (not DI) for a minute or so
- Rinse the beaker out with small amounts of DI water with some swirling. Do this 2-3 times. Place the beaker where it belongs. It is not essential to dry it out with a paper towel
- The eudiometer tube should be filled up a couple of times with DI water and the rinse discarded in sink.

Example Data Analysis

- Use the mass of Mg ribbon to calculate the theoretical yield of H_2 gas
- Calculate the pressure of H_2 gas using the atmospheric pressure and the vapor pressure of H_2O . Convert the pressure to atm units
- Use the $PV=nRT$ equation to calculate R .
- Calculate a percent error
- Do not determine significant digits until the final calculation: record all the digits from your calculator for intermediate calculations

Example Data Analysis

- These are all EXAMPLE DATA, but probably close to what your data will look like

DATA

Mass of the magnesium ribbon 0.04684 g

Temperature of water 21.2°C

Pressure in the room 753.1 mmHg

Vapor pressure of water at your temperature* 18.88 mmHg

Volume of gas in collection tube 42.50 mL

Example Data Analysis

CALCULATIONS

Show your work, complete with units.

1. Moles of hydrogen gas based on mass of Mg used 0.001928 mol H_2

$$0.04686 \text{ g Mg} \times \frac{1 \text{ mol Mg}}{24.31 \text{ g Mg}} \times \frac{1 \text{ mol H}_2 \text{ produced}}{1 \text{ mol Mg consumed}} \\ = 1.928 \times 10^{-3} \text{ mol H}_2 = 0.001928 \text{ mol H}_2$$

2. Pressure of hydrogen gas (mmHg) 734.6 mmHg

$$P_{\text{air}} = P_{\text{water_vapor}} + P_{\text{hydrogen (H}_2\text{)}}$$

$$P_{\text{H}_2} = P_{\text{air}} - P_{\text{water_vapor}} = 753.1 - 18.88 = 734.6 \text{ mmHg}$$

3. Pressure of hydrogen gas (atm) 0.9666 atm H_2

$$734.6 \text{ mmHg} \times \frac{1 \text{ atm}}{760 \text{ mmHg}} = 0.9666 \text{ atm H}_2$$

Example Data Analysis

4. Experimentally determined R value 0.07243 L atm/mol K

$$R = PV/nT$$

$$\begin{aligned} &= 0.9666 \text{ atm} \times 42.50 \text{ mL} \times \frac{1 \text{ L}}{1000 \text{ mL}} \times \frac{1}{0.001928 \text{ mol}} \times \\ &\frac{1}{(273 + 21.2)\text{K}} = \frac{0.07243 \text{ L atm}}{\text{mol K}} \end{aligned}$$

5. % Error -11.73%

$$\frac{(0.07243 - 0.08205)}{0.08205} \times 100\% = -11.73\%$$

6. What measured value would be directly affected if the magnesium ribbon was not cleaned well?

The predicted moles of H_2 gas produced depends on moles of Mg consumed. The moles of Mg consumed will be determined by the mass (grams) of pure Mg solid ribbon, determined by the recording of the mass of ribbon the balance. If the ribbon has Mg compounds (MgO , MgS), these are unreactive and will not contribute to formation of H_2 gas, yet they add to the mass read on the balance.

This measured value would be (too high, too low, unchanged), which will cause the experimental R value to be (higher, lower, no change).

The added mass with impurities predicts a higher amount of H_2 produced than would be produced.

This means that the value 'n' for the ideal gas equation is higher than it should be. A higher value of 'n' would result in a lower value for R constant than it should be ($R = PV/nT$).

7. What measured value would be directly affected if the gas collection tube was not fully filled with water and air remained inside the tube as the tube was inverted?

If the inverted eudiometer was not properly filled with water and had a starting air pocket, that means the measured value which is the volume V for the ideal gas law equation would be larger/higher than it should.

This measured value would be (too high, too low, unchanged), which will cause the experimental R value to be (higher, lower, no change).

The volume V measured quantity is a direct factor in the ideal gas equation that is used to calculate the R gas constant. If the V value is higher than it should be because of an air pocket at start of experiment, then the calculate R constant will be higher than it should

8. Discuss your percent error. Based on your observations, what is at least one source of error specific to your experiment? (Mention the numerical value for your percent error in your response)

In watching the students do the setup of the experiment, I notice they all seem to have no air pocket and they inverted the tube correctly: so volume V value should be correct. Another source of error is not reading the barometric pressure correctly, affecting the P value in the calculation. But this is not likely to be a significant factor. The students did have to record the temperature, which they did dutifully: there is an assumption that the water temperature and the air in the eudiometer do not have a variance in temperature, which might occur and be a source of error. Perhaps the major source of error would be in the mass reading of Mg ribbon: if the ribbon was not cleaned properly and the cleaned ribbon mass not read accurately in the balance, that could contribute to error. Since P, V, T and n are four variables used in the calculation of R, their errors are propagated into the calculation of R.